

Question

Does adding salt to water have any effect on how fast it evaporates (turns from liquid into vapor)?

Background research

Saltwater's freezing point is lower than 0°C (the normal freezing point of pure water) because the dissolved salt gets in the way of the water molecules, making it harder for them to form into solid ice crystals.

Hypothesis

Salt makes it harder for water to form ice, lowering the freezing point. Therefore, does salt also lower the boiling point of water, making it easier to form water vapor? If so, saltwater will evaporate faster than freshwater.

Test the hypothesis

Divide some freshwater into two cups. Add some salt to one cup to make a salt solution. Weigh out 5 ml (0.17 fl oz) of each liquid and pour each amount into two identical shallow dishes. The water should be about 1 mm (0.04 in) deep. Leave the dishes in direct sunlight. Monitor them over a few hours to see which dish dries out first. The hypothesis predicts that the saltwater will evaporate first.

Results

The freshwater dish dries out first. What is the conclusion? Is the hypothesis false or not false?

Conclusion

The hypothesis is false.
Salt in the water does not make it evaporate faster.

